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Giuseppe Silano

Curriculum Vitae

Education

Nov 2020 **Ph.D. in Information Technologies for Engineering**, *Group for Research on Automatic Control Engineering (GRACE)*, University of Sannio, Benevento, Italy
- Additional mention of Doctor Europaeus.
- Focus on robotics, control, path planning, and software-in-the-loop.
- Supervisor: Prof. Dr. Luigi Iannelli.

Mar 2019 – Nov 2019 **Visiting Ph.D. student at Laboratoire d'Analyse et d'Architecture des Systèmes (LAAS), Robotics and Interactions (RIS) group**, *Centre National de la Recherche Scientifique (CNRS)*, Toulouse, France
- Control of fully-actuated 6-DoF[1] robots with onboard sensors.
- Supervisor: Prof. Dr. Antonio Franchi.

Mar 2016 **Master of Science in Electronic Engineering**, *University of Sannio*, Benevento, Italy
- Focus on robotics, control, electronics, and telecommunications.

Jul 2012 **Bachelor of Science in Computer Engineering**, *University of Sannio*, Benevento, Italy
- Focus on robotics, control, software, telecommunications, and electronics.

Feb 2012 – Jun 2012 **Industrial Internship in Systems Engineering**, *Mosaico Monitoraggio Integrato S.r.l.*, Benevento, Italy
- Design of a software production methodology using an Object Oriented (OO) approach for Programmable Logic Controllers (PLCs).

Professional Affiliation

Dec 2016 – Today **IEEE (Institute of Electrical and Electronics Engineers)**, *Student Member (S'17), Member (M'21), Senior Member (SM'25)*
- IEEE Control Systems Society (CSS).
- IEEE Robotics and Automation Society (RAS).
- IEEE Systems, Man, and Cybernetics Society (SMC).

[1] DoFs: Degrees of Freedom.

Academic Appointments

Jul 2022 – Today **Tenured Researcher**, *Department of Power Generation Technologies and Materials (TGM)*, Ricerca sul Sistema Energetico (RSE) S.p.A., Milan, Italy [RP2], [RP3], [RP4], [RP5]

Jul 2022 – Today **Associate Researcher**, *Multi-Robot System (MRS) Group*, Czech Technical University (CTU) in Prague, Prague, Czech Republic [RP6]
- Working with: Prof. Dr. Martin Saska.

Jun 2020 – Today **External Collaborator**, *Group for Research on Automatic Control Engineering (GRACE)*, University of Sannio (UniSannio), Benevento, Italy [RP7]
- Working with: Prof. Dr. Luigi Iannelli.

Jun 2020 – Jul 2022 **Postdoctoral Research Fellow**, *MRS Group*, Czech Technical University in Prague, Prague, Czech Republic [RP6], [M1]
- Supervisor: Prof. Dr. Martin Saska.

Dec 2019 – Jun 2020 **Research Fellow**, *Group for Research on Automatic Control Engineering (GRACE)*, University of Sannio, Benevento, Italy [RP7], [RP9]
- Supervisor: Prof. Dr. Luigi Iannelli.

May 2016 – Nov 2016 **Research Fellow**, *Group for Research on Automatic Control Engineering (GRACE)*, University of Sannio, Benevento, Italy
- Supervisor: Prof. Dr. Luigi Iannelli.

Professional Experience

Jun 2014 – Dec 2025 **Technical Writer**, *Writer of technical articles for magazines and web platforms*, e.g., "`Win Magazine`", "`Ridble`", "`Firmware`", "`EOS Book`", and "`Fare Elettronica`", Italy

Apr 2016 **Junior Software Engineer**, *Software Engine S.r.l.*, Solofra, Italy
- Web Developer (specialized in front-end development) responsible for the maintenance, debugging, and implementation of HTML5 and CSS3 templates (following mobile-first, responsive, and cross-browser principles), JavaScript (jQuery, AngularJS), and back-end integration (Java), as well as the management, implementation, and debugging of relational databases.
- Projects completed:
  - *Software developer for a management system*. Development of software for managing documents submitted under Law 219/81 and subsequent amendments for the town of Mirabella Eclano, Italy.

Aug 2012 – Jan 2013 **Control System Integrator**, *Mosaico Monitoraggio Integrato S.r.l.*, Benevento, Italy
- Design and implementation of control systems for industrial automation based on PLCs and Supervisory Control And Data Acquisition (SCADA).
- Projects completed:
  - *Storage and loading system of soda autoclaves in autoclaving*. Design and development of the system, and analysis of the control system requirements in accordance with the safety functions.
  - *Leaching process of blade cores for high-performance turbines*. Design and development of the system, and analysis of the control system requirements in accordance with the safety functions.

Miscellaneous

Research Areas — Robotics; Uncrewed Aerial Vehicles (UAVs) control and navigation; software- and simulation-in-the-loop; Temporal Logic (TL); robot manipulators; Model Predictive Control (MPC); communication-aware robotics; human-robot collaboration; formal methods in robotics and automation; autonomous systems; middleware for laboratory research infrastructures.

Skills and expertise — **Programming languages**: C, C++, Java, Python. **Miscellaneous**: Jekyll, CMake, MATLAB/Simulink. **Robotics middleware and 3D dynamic simulation environments**: Robot Operating System (ROS), Generator of Modules (GenoM), Gazebo, CoppeliaSim (formerly V-REP), MATLAB Virtual Reality Toolbox. **Markup languages**: Markdown, reStructuredText (RST), LaTeX. **Distributed version control systems**: Mercurial, Git. **Continuous integration systems**: Jenkins, GitHub Actions, TravisCI. **Optimization tools**: Gurobi, CasADi, CVX, IPOPT, qpOASES, YALMIP, MATMPC, Optimization Engine (OpEn), CPLEX. **Containerization platforms**: Docker.

License — **Remote Pilot License**, Remotely Piloted Aircraft Systems Regulation, VL/Mc class, Aeroclub of Benevento Gen. Nicola Collarile, Benevento, Italy, April 2017

Open-source Dev. — Contributor, author, and maintainer of open-source projects:

- One of the authors and maintainers of the TeX package "`europecv`". Comprehensive TeX Archive Network (CTAN) profile available at `https://ctan.org/home/g.silano`.
- Author of the CrazyS, BebopS, and MAT-Fly packages.
- Contributor to the styles, RotorS, bebop_autonomy, NMPC strategy for Autonomous Vehicles, Vision-Based Path Following Algorithm, ros-travis-integration, kaobook, multi-robot inspection and maintenance software architecture, PACNav, JRA-3.1-JaNDER-API, and MATMPC packages, templates, and drawing examples in LaTeX.
- GitHub and Gist profiles: `https://github.com/gsilano` and `https://gist.github.com/gsilano`, respectively.

Research Activities

Motion planning algorithms for the inspection of civilian infrastructure with multi-robot systems

Within the AERIAL-CORE European project [RP6][2], path planning algorithms aimed at computing feasible and constrained trajectories for a fleet of UAVs leveraging Signal Temporal Logic (STL) specifications have been investigated. The planner allows formulating rather complex missions while avoiding obstacles and forbidden areas along the path and accounting for vehicle constraints (e.g., maximum velocity and acceleration) [W6], [W10][2], [J9], and [J13]. Besides, a multi-layer software architecture to perform cooperative missions with a fleet of UAVs was designed [C18], [J10], and several further contributions stemmed from the research activity in [RP6], such as [C12], [C13], [C15], [W5], [W9], [J6], [W4], and [J5]. A thorough analysis of existing robotics simulators was also conducted [J8].

[2]The research activity was partially carried out also within the Comp4Drones European project [RP7].

Communication-aware robotics

The research activity focuses on "*communication-aware robotics*", which addresses the challenges of maintaining connectivity and robust information flow in robotic networks. With the increasing importance of robotic networks in various applications, such as emergency response and surveillance, wireless communication plays a crucial role in connecting UAVs with other nodes or remote operators. The main motivation behind this research is to develop methods and algorithms that enable UAVs to explicitly consider communication link quality during control and motion planning. In this context, several frameworks were designed to generate feasible dynamic trajectories for multi-rotor UAV relays, accounting for both their motion dynamics and the effects induced by changes in antenna orientation [C3], [C4], [C10], [C16]. The presence of jammers and eavesdroppers was also considered [J1], [J2], [W3], [J7], [C1], [C5], [C8].

Model Predictive Control strategies for motion control of aerial robots

Within the AERIAL-CORE European project [RP6], Model Predictive Control (MPC) strategies for vision-based target tracking and collision avoidance in multi-rotor aerial vehicles have been investigated. The proposed control strategies take into account realistic actuation limits, visual perception, and obstacle avoidance constraints [C14]. The research also addressed robust control of cooperative aerial structures under parametric uncertainty [C2], [W1].

Aerial robots formation control

Within the AERIAL-CORE European project [RP6], research activities focused on approaches for achieving decentralized collective navigation of UAV swarms. Simulated and real-world experiments conducted in a natural forest demonstrated the validity and effectiveness of the proposed collective navigation approach in challenging environments [J11], [W5]. Additionally, a comprehensive analysis of existing robotics simulators was carried out [J8].

Modeling and control of autonomous vehicles

An MPC strategy aimed at controlling a small-scale car model for autonomous racing competitions has been investigated. An optimization problem was set up considering both the vehicle's actuation limits and the lateral and longitudinal forces acting on the car, described by Pacejka's Magic Formula and a drivetrain model. The approach allows the vehicle to race safely on a track populated by static obstacles, generating collision-free trajectories and tracking them while improving lap-time performance. The feasibility and effectiveness of the control architecture are demonstrated through closed-loop simulations in Gazebo, specifically using the F1/10 simulator [C11][2].

Electric Vehicles Integration into Smart Grids

Within the FLOW European project [RP4], the research focused on developing innovative strategies for integrating Electric Vehicles (EVs) into the

power grid to improve flexibility and efficiency. The work aims to optimize EV deployment, promote sustainable energy management, and advance the seamless integration of EVs within smart grid infrastructures [J4].

Control allocation and actuator readiness

Building on the study of redundancy resolution for overactuated aerial platforms[3], research has focused on treating control authority as a certified, forward-invariant resource rather than a byproduct of the allocation objective. A control barrier function on the log-determinant of a drag-aware actuator-authority metric was designed to reconcile readiness-maximizing and effort-minimizing allocators, enforcing a certified authority floor at negligible tracking cost for fixed-geometry multirotors [P3]. This framework was then extended to articulated, tilting-rotor platforms, where servo tilt is exploited as an active-morphology resource to preserve motor wrench-rate authority when fixed-geometry designs face a zero-sum trade-off between authority and tracking, with the resulting certificate enforced through a unified physical-command quadratic program acting jointly on motor torques and servo setpoints [P1], [P2].

Physical human-aerial robot interaction

Robots are designed to assist humans in physically demanding tasks and enhance their mental and physical well-being. Collaboration between humans and robots is crucial for tackling complex tasks, particularly in unstructured environments. However, human-aerial robot interaction remains largely unexplored due to the potential dangers that aerial vehicles pose to humans. To address this, a method has been developed that uses high-level specifications expressed in the form of an Signal Temporal Logic (STL) formula to automate task assignments and trajectories for a multi-rotor aerial vehicle performing object handovers in a power line setting. This method also considers human comfort factors such as ergonomics and preferences [J3], [C9], [W2], [W8][4], [A1]. Additionally, multi-UAV formations have been investigated to further assist human workers in such critical environments [J5], [W7]. A comprehensive analysis of existing robotics simulators was also conducted [J8].

Control algorithms for autonomously managing production and coordinating agricultural operations

Within the AFarCloud European project [RP9], control techniques aimed at autonomously managing production and coordinating farming operations have been investigated, i.e., hierarchical planning algorithms (e.g., RRT, A*, etc.). Specifically, algorithms were developed to coordinate operations within autonomous farming activities, enhancing productivity and reducing costs while gradually shifting towards a high level of autonomy [C17][2], [C20], and [J12][2].

Flight control of Uncrewed Aerial Vehicles

Various activities dealing with UAVs, in particular multi-rotor vehicles, [J14][2], [BC4], [C17][2], [C19], [C21], and [C22] have been carried out.

[3]See 10.1109/LRA.2026.3723738, 10.1109/ICUAS69441.2026.11598607, etc.

[4]The research activity was partially carried out within the research visit spent at the University of Twente and funded by the European H2020 TERRINet project (grant agreement no. 730994), call no. 10.

Software-in-the-loop (SIL) simulation has been investigated as an effective tool for analyzing the behavior of the closed control loop [BC4] and for helping in the design of control strategies [C19].

- **CrazyS**: CrazyS is an extension of the ROS package RotorS, aimed at modeling, developing, and integrating the Crazyflie 2.0 nano quad-rotor in the physics-based simulation environment Gazebo. This simulation platform allows quickly understanding the behavior of the flight control system by comparing and evaluating different indoor and outdoor scenarios, with a level of detail close to reality. The proposed extension was developed using Ubuntu 16.04 and the Kinetic Kame version of ROS, but it is also fully compatible with Ubuntu 18.04 along with the Melodic Morenia distribution of ROS and Ubuntu 20.04 along with the Noetic Ninjemys release of ROS. The platform expands the RotorS capabilities by considering the Crazyflie 2.0 physical model, its flight control system, and the Crazyflie's onboard Inertial Measurement Unit (IMU). The code has been released as open-source under the Apache license (`https://github.com/gsilano/CrazyS` with the aim of sharing the obtained results with other researchers who already use such tools and would like to use the platform [BC4], [C21].)
- **BebopS**: The repository contains the ROS code developed for the "Aerial robotics control and perception challenge", the Industrial Challenge of the 26th Mediterranean Conference on Control and Automation (MED'18) [A4]. The code aims to simulate the dynamics of the Parrot Bebop 2 together with the flight controller (both high and low level) when external disturbances (e.g., wind gusts) act on it. The control algorithms were designed in MATLAB/Simulink and validated in Gazebo by using the MathWorks Robotics System Toolbox (MRST). Therefore, the repository contains both the ROS nodes and the launch files needed to simulate the drone behavior when a tracking algorithm is run. The glue code written during the challenge to tackle the tracking problem with the real platform is also available on GitHub. Such code allows sending control commands and receiving data from the aircraft and the Motion Capture (MoCap) system, i.e., Vicon. Finally, the repository contains the developed ROS code to connect the control algorithm to the Parrot Sphinx simulator, including the Parrot Bebop onboard firmware in the loop. In this way, details such as synchronization, timing issues, fixed-point computation, overflow, and divisions by zero can be isolated and investigated in detail [C19]. The code has been released as open-source under the Apache license at `https://github.com/gsilano/BebopS`.

- **MAT-Fly**: MAT-Fly is a numerical simulation platform for multi-rotor aircraft characterized by ease of use and of control development. The platform is based on MATLAB and the MathWorks Virtual Reality Toolbox (MVRT), which work together to simulate the behavior of a drone in a 3D environment while tracking a car moving along a nontrivial path. The MVRT has been chosen due to the familiarity that students have with MATLAB and because its simple structure allows shifting the attention to the classifier, the tracker, the reference generator, and the trajectory tracking controller. The overall architecture is modular, so that each block can be easily replaced with others, simplifying the development phase and allowing further functionalities to be added. The code is released under the Apache license and made available at `https://github.com/gsilano/MAT-Fly` and `https://www.mathworks.com/matlabcentral/fileexchange/72486-mat-fly`, thus making the numerical platform available for scientific and educational activities [J14], [C22].

Mohamed Bin Zayed International Robotics Challenge 2020

The Mohamed Bin Zayed International Robotics Challenge (MBZIRC) is a biennial international robotics competition that provides an ambitious and technologically demanding set of challenges and is open to teams from all countries. The MBZIRC competition aims to inspire the future of robotics through innovative solutions and technological excellence. Within the competition, specifically "Challenge 2" of the 2020 edition, autonomous aerial and ground robots were developed for navigation and manipulation tasks in unstructured outdoor and indoor environments. In particular, a team of UAVs and a Uncrewed Ground Vehicle (UGV) collaborated to autonomously locate, pick, transport, and assemble different types of brick-shaped objects to build predefined structures in an outdoor environment. The challenge is motivated by construction automation and autonomous robot-based 3D printing of large structures. This research activity was carried out during the visiting period spent at LAAS-CNRS in Toulouse [A3], [W11].

Services for research infrastructure integration, coupling, and automation

A driving force behind the realization of a sustainable energy supply is the integration of renewable energy resources. Due to their stochastic generation behavior, energy utilities are confronted with the challenge of managing more complex power grids. Additionally, technological advancements, controllable loads, integration with other energy sources, changing regulatory rules, and market liberalization require the adaptation of the system's operation. Proper operational concepts and intelligent automation provide the foundation to transform the existing power system into an intelligent entity known as a smart grid. Within the context of the H2020 ERIGrid 2.0 European project [RP3], tools and methodologies have been designed to facilitate the integration, coupling, and automation of renewable energy resources [BC1], [BC2], [BC3], [C6], [C7].

Research Projects

[RP1] **Principal Investigator of a GAČR Junior Star grant**, *Knowledge-State-Driven Aerial Inspection of Electrical Energy Infrastructure*, funded by the Czech Science Foundation (GAČR), grant no. 27-00437M, Hosted by the Multi-Robot Systems Group, Czech Technical University in Prague, Head of the host group: Prof. Martin Saska, Project amount: 25 MCZK (approx. 1.0 MEuros), Role: Principal Investigator, January 2027 – December 2031

[RP2] **Participation within the research fund for the Italian Electrical System**, *Resilience and Security of the Energy System project*, financed under the contract agreement between RSE S.p.A. and the Ministry of the Environment and Energy Security, Project Advisor for Ricerca sul Sistema Energetico (RSE) S.p.A.: Eng. Giovanni Pirovano and Eng. Riccardo Lazzari, Project amount: 18.6 MEuros (only related to RSE activities), Role: leader from RSE's Department side (WP3 – LA3.01, LA3.04, and LA3.07 leader), January 2025 – Today

[RP3] **Participation within H2020-2019-1-RIA - INFRAIA**, *European Research Infrastructure supporting Smart Grid Systems Technology Development, Validation and Roll Out (ERIGRID2)*, research project agreement no. 870620, Project Advisor for Ricerca sul Sistema Energetico (RSE) S.p.A.: Eng. Riccardo Lazzari, Project amount: 561 kEuros (only related to RSE activities), Role: work package leader (WP13 – JRA4), October 2022 – April 2025

[RP4] **Participation within HORIZON-CL5-2021-D5-01-RIA**, *Flexible energy systems Leveraging the Optimal integration of EVs deployment Wave (FLOW)*, research project agreement no. 101056730, Project Advisor for Ricerca sul Sistema Energetico (RSE) S.p.A.: Eng. Filippo Colzi, Project amount: 458 kEuros (only related to RSE activities), Role: leader from RSE's Department side (T4.3 leader), July 2022 – June 2026

[RP5] **Participation within the research fund for the Italian Electrical System**, *Vehicle-to-Grid (V2G) project*, financed under the contract agreement between RSE S.p.A. and the Ministry of Economic Development - General Directorate for the Electricity Market, Renewable Energy and Energy Efficiency, Nuclear Energy, Project Advisor for Ricerca sul Sistema Energetico (RSE) S.p.A.: Eng. Filippo Colzi and Eng. Riccardo Lazzari, Project amount: 8.4 MEuros (only related to RSE activities), Role: leader from RSE's Department side (WP2 – LA2.01, LA2.04, and LA2.07 leader), July 2022 – December 2024

[RP6] **Participation within H2020-2019-2-RIA-two stages - ICT**, *AERIAL COgnitive integrated multi-task Robotic system with Extended operation range and safety (AERIAL-CORE)*, research project agreement no. 871479, Project Advisor for the Czech Technical University in Prague: Prof. Dr. Martin Saska, Project amount: 461 kEuros (only related to CTU activities), Role: leader from CTU's side (T2.3 and T7.4 leader), June 2020 – July 2022, and participant, July 2022 – December 2023

[RP7] **Participation within H2020-2018-2-RIA-two-stages – ECSEL**, *Software Components For Drones (Comp4Drones)*, research project agreement no. 826610, Project Advisor for the University of Sannio: Prof. Dr. Luigi Iannelli, Project amount: 285 kEuros (only related to University of Sannio activities), Role: participant, October 2019 – May 2023

[RP8] **Participation within the Mohamed Bin Zayed International Robotics Challenge (MBZIRC) 2020**, Project Advisor: Prof. Dr. Antonio Franchi, Role: member of the Laboratoire d'Analyse et d'Architecture des Systèmes (LAAS) team, March 2019 – February 2020

[RP9] **Participation within H2020-2017-2-RIA-two-stages – ECSEL**, *Aggregate Farming in the Cloud (AFarCloud)*, research project agreement no. 783221, Project Advisor for the University of Sannio: Prof. Dr. Luigi Glielmo, Role: leader from UniSannio side (T3.2 leader), September 2018 – November 2021

Research Visits

2022 November 17 – November 30, 2022, University of Twente, Enschede, The Netherlands, Robotics and Mechatronics (RaM) group, Prof. Dr. Antonio Franchi, funded by the European H2020 TERRINet project (grant agreement no. 730994), call no. 10.

Scientific Activity[5]

Editorial Board

2026 – 2028 Associate Editor of the IEEE Transactions on Automation Science and Engineering (T-ASE).

2025 – 2027 Associate Editor of the IEEE Control Systems Society (CSS) Conference Editorial Board.

2025 – 2027 Associate Editor of the IEEE International Conference on Robotics & Automation (ICRA).

2022 – 2026 Associate Editor of the International Conference on Unmanned Aircraft Systems (ICUAS).

2024 – 2026 Early Career Editorial Board Member of Biomimetic Intelligence and Robotics (BIRob).

2025 Associate Editor of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

2025 Associate Editor of the IEEE International Conference on Systems, Man, and Cybernetics (SMC).

2024 – 2025 Associate Editor of the Mediterranean Conference on Control and Automation (MED).

2023 – 2024 Member of the Track Program Committee of Robotics: Science and Systems (RSS). Role: reviewer.

[5]Link to the Web of Science profile: `https://www.webofscience.com/wos/author/record/1668893`.

2023 Member of the Track Program Committee of the 2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE'23). Member of the Technical Track Committee "TT 03 - Power Systems and the Smart Grid, Renewable Energy Systems and Smart Grid". Role: reviewer.

Expert Reviewer for Funding Agencies

2026 Remote Evaluator for the European Innovation Council (EIC Transition – Horizon Europe), a flagship EU programme supporting breakthrough innovation.

2025 – 2026 Member of the Register of Evaluation Experts (EIV) for ANVUR (Italian National Agency for the Evaluation of Universities and Research Institutes). Tasked with the peer review of research outputs (e.g., articles, monographs, patents) within the Expert Evaluation Panel (GEV) for GEV 09 - Industrial and Information Engineering, in accordance with the Agency's criteria and guidelines.

2025 Registered expert in the REPRISE register (Register of Scientific Experts) established by the Italian Ministry of University and Research (MUR), under the section Competitive Industrial Research and Social Development.

2025 Reviewer for the Agence Nationale de la Recherche (ANR), one of the leading governmental science funding organizations in France, supporting innovation and quality in research.

2025 Reviewer for the Dutch Research Council (NWO), one of the leading governmental science funding organizations in the Netherlands, supporting innovation and quality in research.

Scientific Societies

2023 – Today Member of the IEEE Robotics and Automation Society (RAS) Technical Committee on Smart Buildings

2023 – Today Member of the IEEE Robotics and Automation Society (RAS) Technical Committee on Algorithms for Planning and Control of Robot Motion

2023 – Today Member of the IEEE Robotics and Automation Society (RAS) Technical Committee on Aerial Robotics and Unmanned Aerial Vehicles

2023 – Today Member of the IEEE Robotics and Automation Society (RAS) Technical Committee on Space Robotics

2022 – Today Member of the IEEE Robotics and Automation Society (RAS) Technical Committee on Multi-Robot Systems

Edited Books Reviewer

2020 Koubaa, Anis (Ed.), "Robot Operating System (ROS) - The Complete Reference (Volume 6)", Springer International Publishing, 2021.

2019 Koubaa, Anis (Ed.), "Robot Operating System (ROS) - The Complete Reference (Volume 5)", Springer International Publishing, 2020.

2018 Koubaa, Anis (Ed.), "Robot Operating System (ROS) - The Complete Reference (Volume 4)", Springer International Publishing, 2019.

International Journals Reviewer

2026 IEEE Transactions on Neural Networks and Learning Systems

2026	Science Robotics
2026	Advanced Robotics Research
2026	MDPI Drones
2026	ICUAS Association Unmanned Aviation eMagazine
2025	IEEE Open Journal of the Communications Society
2025 – Today	IEEE Systems, Man, and Cybernetics Magazine
2025	IEEE Wireless Communications
2025 – Today	The International Journal of Robotics Research (IJRR)
2025	IEEE Communications Letters
2025 – Today	IEEE Robotics & Automation Magazine (RAM)
2025 – Today	IEEE Transactions on Automation Science and Engineering (T-ASE)
2025	IEEE Transactions on Intelligent Transportation Systems (T-ITS)
2025	IEEE Internet of Things Journal (IoT-J)
2025	IEEE/ASME Transactions on Mechatronics (T-MECH)
2024	Biomimetic Intelligence and Robotics
2024	Computational and Structural Biotechnology Journal (CSBJ)
2024 – Today	Engineering Applications of Artificial Intelligence (EAAI)
2024	IEEE Control Systems Letters (L-CSS)
2024	IEEE Open Journal of Control Systems (OJ-CSYS)
2024	IEEE Open Journal of the Industrial Electronics Society (OJIES)
2024	IEEE Open Journal of Vehicular Technology (OJVT)
2024 – Today	Robotics and Autonomous Systems (RAS)
2024 – Today	IEEE Transactions on Industrial Informatics (TII)
2024	IEEE Transactions on Intelligent Vehicles
2024	IEEE Transactions on Systems, Man, and Cybernetics: Systems
2024 – Today	IEEE Transactions on Robotics (T-RO)
2023	Archives of Electrical Engineering
2023 – Today	Control Engineering Practice
2023 – Today	Expert Systems With Applications
2023	ISA Transactions
2023	PLoS ONE
2023	Nature Scientific Reports
2023	Taylor & Francis Cybernetics and Systems
2023	IEEE Transactions on Circuits and Systems for Video Technology
2022 – Today	Automatica
2022 – Today	Field Robotics
2022	IEEE Transactions on Automatic Control (TAC)
2022	IEEE Vehicular Technology Magazine
2022	Journal of Field Robotics
2021	IET Cyber-Systems and Robotics

2021 MDPI Energies

2021 Journal of Unmanned Vehicle Systems

2021 – Today IEEE Access

2021 – Today IEEE Robotics and Automation Letters (RA-L)

2019 – Today IEEE Transactions on Control Systems Technology (TCST)

2019 – Today Springer Journal of Intelligent & Robotic Systems (JINT)

International Conferences Reviewer

2025 IEEE Conference on Advanced Robotics and its Social Impact (ARSO)

2018 – Today IEEE American Control Conference (ACC)

2021 – Today IEEE International Conference on Systems, Man, and Cybernetics (SMC)

2021 – Today IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

2018 – Today IEEE Conference on Control Technology and Applications (CCTA)

2020 – Today International Conference on Unmanned Aircraft Systems (ICUAS)

2023 – Today Mediterranean Conference on Control and Automation (MED)

2021 – Today IEEE International Conference on Robotics and Automation (ICRA)

2020 – Today IEEE International Workshop on Metrology for Aerospace (MAS)

2019 – Today IEEE European Control Conference (ECC)

2019 – Today IEEE Conference on Decision and Control (CDC)

2025 IFAC Symposium on Mechatronic Systems

2024 IEEE TechDefense 2024

2023 International Conference on System Theory, Control and Computing (ICSTCC)

2023 – Today International Workshop on "Open Source Modelling and Simulation of Energy Systems" (OSMSES)

2023 International Federation of Automatic Control (IFAC)

2021 Workshop on Aerial Robotic Systems Physically Interacting with the Environment (AIRPHARO)

Conference Services

Program Committees

2026 **Member of the Program Committee**, *Annual National Meeting of SIDRA (Italian Society of Automatic Control), AUTOMATICA.IT 2026, University of Parma, September 2–4, 2026*, Website: `https://automatica2026.unipr.it/index.php/committees`

2025 **Tutorial Co-Chair**, *2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC), October 5–8, 2025, Vienna, Austria*, Website: `https://www.ieeesmc2025.org/committees.html`

Organized Tutorial Sessions

Jun 2026 **Organizer of the tutorial session "When Robotics Meet Wireless Communications: A Tale of Two Interconnected Worlds"**, *2026 International Conference on Unmanned Aircraft Systems (ICUAS'26), June 15–18, 2026, Corfu, Greece*, together with Prof. Dr. Daniel Bonilla Licea, Prof. Dr. Hajar El Hammouti, and Prof. Dr. Mounir Ghogho, Mohammed VI Polytechnic University, Ben Guerir, Morocco, and Prof. Dr. Martin Saska, Czech Technical University in Prague, Czech Republic. Website: `https://uasconferences.com/2026_icuas/t2-half-day-workshop/`

Oct 2025 **Organizer of the tutorial session "Redefining UAV Wireless Networks: The Role of Fully-Actuated Multi-Rotor Systems in Robotics and Communication"**, *2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC'25), October 5–8, 2025, Vienna, Austria*, together with Prof. Dr. Daniel Bonilla Licea, Prof. Dr. Hajar El Hammouti, and Prof. Dr. Mounir Ghogho, Mohammed VI Polytechnic University, Ben Guerir, Morocco, and Prof. Dr. Martin Saska, Czech Technical University in Prague, Czech Republic. Website: `https://www.ieeesmc2025.org/tutorials.html`

Organized Workshops

May 2023 **Organizer of the workshop session "The Role of Robotics Simulators for Unmanned Aerial Vehicles"**, *2023 IEEE International Conference on Robotics and Automation (ICRA'23), May 29 – June 2, 2023, ExCeL London, UK*, together with Dr. Kimberly McGuire, Bitcraze AB, Malmö, Sweden, Prof. Dr. Chiara Gabellieri, Robotics and Mechatronics group, Faculty of Electrical Engineering, Mathematics & Computer Science, University of Twente in Enschede, The Netherlands, and Prof. Dr. Wolfgang Hönig, Intelligent Multi-Robot Coordination Lab of TU Berlin, Germany. Website: `https://imrclab.github.io/workshop-uav-sims-icra2023`

Organized Special Issues

Jul 2023 – Oct 2025 **Organizer of the special issue "Advanced Control and Path Planning of Unmanned Aerial Vehicles (UAVs)"**, *in MDPI Machines (ISSN 2075-1702), section "Mechatronic and Intelligent Machines"*, together with Prof. Dr. Yahui Liu, School of Vehicle and Mobility, Tsinghua University, Beijing, China. Website: `https://www.mdpi.com/journal/machines/special_issues/7ZS283VVX8`

Organized Invited Sessions

Jun 2024 **Organizer of the invited session "Aerial Robotics in Inspection and Maintenance Operations - From Design to Deployment"**, *2024 International Conference on Unmanned Aircraft Systems (ICUAS), June 4–7, 2024, Chania, Crete, Greece*, together with Prof. Dr. Chiara Gabellieri, Robotics and Mechatronics group, Faculty of Electrical Engineering, Mathematics & Computer Science, University of Twente in Enschede, The Netherlands, and Prof. Dr. Mario Selvaggio, Department of Electrical Engineering and Information Technology, University of Naples Federico II, Naples, Italy. Website: `https://sites.google.com/view/icuas-2024-special-session/home`

Invited Talks

Nov 2024 **Exploring Laboratory Middleware for Distributed Simulations Across Research Infrastructures: A Case Study from the ERIGrid 2.0 Project**, *Invited Talk*, The 14th Mediterranean Conference on Power Generation, Transmission, Distribution and Energy Conversion (MED POWER 2024). Panel no. 4 "Advanced simulation and laboratory methods in power systems", Athens, Greece, `https://medpower2024.org/panels/`

Sep 2024 **Are Robotic Simulators a Solved Problem? Criteria for Selection and Customization**, *Invited Seminar*, ROS Aerial Community, Online Meeting, `https://discourse.ros.org/t/september-2024-meetings-aerial-robotics-back-from-summer/39383`

Dec 2022 **Signal Temporal Logics as a Tool to Encode Complex Missions for Multi-Robot Aerial Systems**, *Invited Seminar*, University of Sannio, Benevento, Italy

Apr 2016 **Advanced drone applications: opportunities and problems**, *Invited Talk*, Make & Share event, Tech Coffee association, Benevento, Italy

Session Chair or Co-Chair at International Conferences

Oct 2025 2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC'25), Vienna, Austria. Role: chair.

Jun 2021 2021 International Conference on Unmanned Aircraft Systems (ICUAS'21), Athens, Greece. Role: chair.

Conference Contributions

Sep 2024 **2024 Open Source Modelling and Simulation of Energy Systems (OSMSES'24)**, *"Integrating Power-to-Heat Services in Geographically Distributed Multi-Energy Systems: A Case Study from the ERIGrid 2.0 Project"*, Vienna, Austria, Oral Presentation

May 2024 **2024 12th Workshop on Modelling and Simulation of Cyber-Physical Energy Systems (MSCPES'24)**, *"Laboratory Middleware for the Cyber-Physical Integration of Energy Research Infrastructures"*, Hong Kong, China (Virtual), Oral Presentation

Oct 2022 **2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC'22)**, *"A Nonlinear Model Predictive Control Strategy for Autonomous Racing of Scale Vehicles"*, Prague, Czech Republic, Oral Presentation

Sep 2021 **Automatica.it 2021**, *national meeting held by SIDRA[6], "Optimum Trajectory Planning for Multi-Rotor UAV Relays with Tilt and Antenna Orientation Variations"*, Catania, Italy, Oral Presentation

Jun 2021 **2021 International Conference on Unmanned Aircraft Systems (ICUAS'21)**, *"A Multi-Layer Software Architecture for Aerial Cognitive Multi-Robot Systems in Power Line Inspection Tasks"*, Athens, Greece, Oral Presentation

Jun 2021 **2021 IEEE International Conference on Robotics and Automation (ICRA'21)**, *"Power Line Inspection Tasks with Multi-Aerial Robot Systems via Signal Temporal Logic Specifications"*, Xi'an, China, Oral Presentation

[6]SIDRA: Italian Society of Automatic Control.

Sep 2020 **Automatica.it 2020**, *national meeting held by SIDRA[6], "A framework for power line inspection tasks with multi-robot systems from signal temporal logic specifications"*, Cagliari, Italy, Oral Presentation

Jun 2018 **26th Mediterranean Conference on Control and Automation (MED'18)**, *"CrazyS: a software-in-the-loop platform for the Crazyflie 2.0 nano-quadcopter"*, Zadar, Croatia, Oral Presentation

Sep 2017 **Automatica.it 2017**, *national meeting held by SIDRA[6], "An educational simulation platform for Unmanned Aerial Vehicles aimed to detect and track moving objects"*, Milan, Italy, Poster Session

Teaching Experience

Teaching Activity

Jul 2024 **Teaching Activity for the Ph.D. course "Fundamentals for Robot Programming with ROS" (1 ECTS credit)**, *Ph.D. programme in Information Technologies for Engineering, Department of Engineering, University of Sannio*, Benevento, Italy, Academic Year 2023/2024

Teaching Assistance

Sep 2017 – Dec 2019 **Teaching Assistant for the graduate course "Discrete Systems" (9 ECTS credits)**, *first year of the Master's degree (Laurea Magistrale) in Computer Engineering, Department of Engineering, University of Sannio*, Benevento, Italy, Academic Years 2017/2018 and 2018/2019, Supervisor: Prof. Dr. Luigi Iannelli

Mar 2016 – Jun 2019 **Teaching Assistant for the undergraduate course "Automatic Control" (6 ECTS credits)**, *second year of the Bachelor's degree (Laurea Triennale) in Computer and Electronics Engineering, Department of Engineering, University of Sannio*, Benevento, Italy, Academic Years from 2016/2017 to 2018/2019, Supervisor: Prof. Dr. Luigi Iannelli

Sep 2018 – Dec 2018 **Teaching Assistant for the graduate course "Advanced Controls" (9 ECTS credits)**, *second year of the Master's degree (Laurea Magistrale) in Electronics Engineering, Department of Engineering, University of Sannio*, Benevento, Italy, Academic Year 2018/2019, Supervisor: Prof. Dr. Luigi Glielmo

Subject Matter Expert ("Cultore della materia")

2017 – 2019 **Subject Matter Expert ("Cultore della materia")**, *University of Sannio*, Benevento, Italy, expert for the course "Sistemi Discreti" (art. 46 D.P.R. 28/12/2000 n. 445 of Italian law), under the supervision of Prof. Dr. Luigi Iannelli, Master's degree in Computer Engineering

2017 – 2018 **Subject Matter Expert ("Cultore della materia")**, *University of Sannio*, Benevento, Italy, expert for the course "Controlli Automatici" (art. 46 D.P.R. 28/12/2000 n. 445 of Italian law), under the supervision of Prof. Dr. Luigi Iannelli, Bachelor's degree in Electronics Engineering

Co-supervisor Scientific Activity

2020 **MIT Independent Activities Period project of Ria Sonecha**, *"Software- and hardware-in-the-loop methodologies aimed to test and validate the open-source simulation platform CrazyS"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2018 **MIT Independent Activities Period project of Benjamin Rodriguez**, *"Crazyflie 2.0 model and control for Gazebo 3D simulator"*, University of Sannio, Benevento, Italy, Advisors: Prof. Dr. Luigi Glielmo, Prof. Dr. Luigi Iannelli, and Prof. Dr. Davide Liuzza

Student Supervisor

2021 **Bachelor Thesis of Andriy Dmytruk**, *"A perception-aware NMPC for collision avoidance and control of a multi-rotor UAV with visual localization constraints"*, Ukrainian Catholic University, Ukraine, Advisor: Dr. Giuseppe Silano

2021 **Master Project Thesis of Giuseppe Caputo**, *"A framework to fulfill inspection missions for a multi-robot system via Linear Temporal Logic specifications"*, Czech Technical University in Prague, Czech Republic, Advisor: Dr. Giuseppe Silano

2021 **Master Project Thesis of Antonio Lombardo**, *"Signal Temporal Logic as a tool to encode inspection missions for a quad-rotor"*, Czech Technical University in Prague, Czech Republic, Advisor: Dr. Giuseppe Silano

Student Co-supervisor

2017 **Bachelor Thesis of Emanuele Aucone**, *"Hovering control for a nano quad-rotor"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2017 **Bachelor Thesis of Francesco Mariano D'Andrea**, *"Adaptive control of the vertical dynamics of a drone"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2017 **Bachelor Thesis of Olga Napolitano**, *"State estimation for a quad-rotor"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2017 **Bachelor Thesis of Susanna Maio**, *"Analysis of the vertical control of a quad-rotor in the V-REP simulation environment"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2016 **Bachelor Thesis of Andrea Mascia**, *"Study of the algorithm for the position estimation of a micro UAV"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

2016 **Bachelor Thesis of Luca De Vincentis**, *"Experimental characterization of the flight control system of a micro UAV"*, University of Sannio, Benevento, Italy, Advisor: Prof. Dr. Luigi Iannelli

Awards

[A1] **Winner of the Best Extended Abstract Award of the IEEE RAS TC on Aerial Robotics and Unmanned Aerial Vehicles (ARUAVs)**, *Workshop "Advancements in Aerial Physical Interaction" at the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2025*, Paper: "STL-Guided Trajectory Planning with Uncertainty-Aware Risk Assessment for Human–Aerial Robot Collaboration", USD 500 travel grant

[A2] **Among the participants of the UNISANNIO team, winner of the "MathWorks Minidrone Competition at IFAC2020"**, *International Robotics Competition*, organized by MathWorks, Berlin, Germany, Webpage: `https://bit.ly/2VLZZU0`, `http://tiny.cc/46p3pz`

[A3] **Among the participants of the LAAS team [RP8], finalist of the "Mohamed Bin Zayed International Robotics Challenge (MBZIRC)"**, *International Robotics Competition*, organized by Khalifa University, Abu Dhabi, United Arab Emirates, Webpage: `http://mbzirc2020team.laas.fr/`, `http://mbzirc.com/`, Video: `https://youtu.be/0uWvDba7Eug`

[A4] **Finalist of the "Aerial robotics control and perception challenge"**, *26th Mediterranean Conference on Control and Automation (MED'18) Industrial Challenge*, organized by the University of Zagreb, Croatia, ISSN: 1066-033X, DOI: `10.1109/MCS.2018.2888719`, URL: `http://dx.doi.org/10.1109/MCS.2018.2888719`

References

Prof. Dr. Luigi Iannelli, Department of Engineering, University of Sannio, Piazza Roma 21, 82100 Benevento, Italy, luigi.iannelli@unisannio.it

Prof. Dr. Martin Saska, Faculty of Electrical Engineering (FEL), Czech Technical University in Prague, Karlovo namesti 13, 12135, Prague 2, Czech Republic, martin.saska@cvut.cz

Prof. Dr. Antonio Franchi, Faculty of Electrical Engineering, Mathematics & Computer Science, University of Twente, Carré 3609, P.O. Box 217, 7500 AE Enschede, The Netherlands, a.franchi@utwente.nl

Prof. Dr. Davide Liuzza, Department of Engineering, University of Sannio, Piazza Roma 21, 82100 Benevento, Italy, davide.liuzza@unisannio.it

Prof. Dr. Luigi Glielmo, Department of Electrical Engineering and Information Technology, University of Naples Federico II, Via Claudio 21, 80125 Naples, Italy, luigi.glielmo@unina.it

Prof. Dr. Wolfgang Hönig, Institute of Computer Engineering and Microelectronics, EECS Faculty, TU Berlin, Marchstraße 23, 10587 Berlin, Germany, hoenig@tu-berlin.de

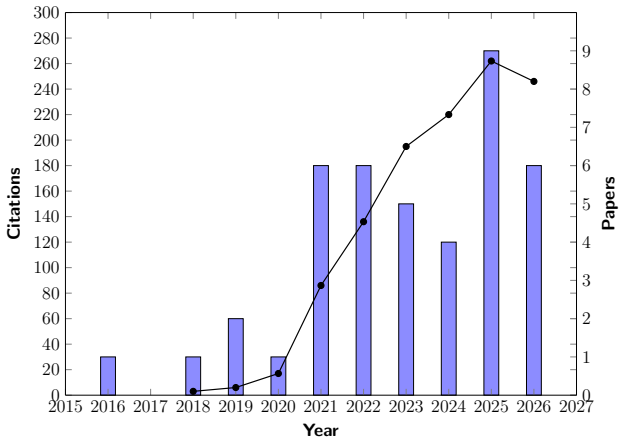
Prof. Dr. Mounir Ghogho, Mohammed VI Polytechnic University, Rabat Campus, Rocade Rabat-Salé, 11103 Rabat, Morocco, mounir.ghogho@um6p.ma

Eng. Carlo Sandroni, Department of Power Generation Technologies and Materials, Ricerca sul Sistema Energetico, Via Raffaele Rubattino 54, 20134 Milan, Italy, carlo.sandroni@rse-web.it

Eng. Riccardo Lazzari, Department of Power Generation Technologies and Materials, Ricerca sul Sistema Energetico, Via Raffaele Rubattino 54, 20134 Milan, Italy, riccardo.lazzari@rse-web.it

Publications

The plot illustrates the progression of citations and the number of papers published over time, while the table below provides a summary of key citation metrics derived from Google Scholar. Additionally, I have had the privilege of **collaborating with over 86 unique co-authors**, with **17 recurring collaborators across ≥ 3 papers**, throughout my research career.



Citations Summary	
Total Citations	1185
h-index	16
i10-index	22

Publication Category	
Journal Articles	14
Conference Papers	22
Book Chapters	4
Workshop Papers	11
Preprints	3

Preprints

[P1] **G. Silano**. "*Delay-Certified Readiness Barriers for Overactuated Multirotor Allocation Without Lipschitz Constants*", November 2026, arXiv, DOI: 10.48550/arXiv.2611.03490, URL: https://doi.org/10.48550/arXiv.2611.03490.

[P2] **G. Silano** and M. Saska. "*Tilt as a Certified Resource: Preserving Motor Wrench-Rate Authority on Articulated Multirotors*", September 2026, arXiv, DOI: 10.48550/arXiv.2609.21580, URL: https://doi.org/10.48550/arXiv.2609.21580.

[P3] **G. Silano**. "*Readiness Barrier Functions: Forward-Invariant Control Authority for Overactuated Multirotor Allocation*", August 2026, arXiv, DOI: 10.48550/arXiv.2608.16335, URL: http://arxiv.org/abs/2608.16335.

Referred Journals

[J1] **G. Silano**, D. Bonilla Licea, D. Liuzza, A. Franchi, and M. Saska. "*Communications-Aware NMPC for Multi-Rotor Aerial Relay Networks Under Jamming Interference*", IEEE Transactions on Vehicular Technology, early access, 2026, DOI: 10.1109/TVT.2026.3738530, URL: http://doi.org/10.1109/TVT.2026.3738530. Impact Factor: 7.4.

[J2] A. K. A. H. Saliah, H. El Hammouti, D. Bonilla Licea, and **G. Silano**. "*Orientation-Aware Control and Trajectory Design for Aerial RIS-Assisted Wireless Communications*", IEEE Transactions on Vehicular Technology, early access, 2026, DOI: 10.1109/TVT.2026.3736485, URL: http://doi.org/10.1109/TVT.2026.3736485. IMPACT FACTOR: 7.4.

[J3] **G. Silano**, A. Afifi, M. Saska, and A. Franchi. "*STL-Based Motion Planning and Uncertainty-Aware Risk Analysis for Human-Robot Collaboration with a Multi-Rotor Aerial Vehicle*", Journal of Intelligent & Robotic Systems, vol. 112, no. 66, pp. 1-31, July 2026, DOI: 10.1007/s10846-026-02403-y, URL: http://doi.org/10.1007/s10846-026-02403-y. IMPACT FACTOR: 3.1.

[J4] D. Liuzza, E. Mostacciuolo, V. Mariani, L. Iannelli, F. Vasca, and **G. Silano**. "*Optimal Online Management of Vehicle-to-Grid Charging in Corporate Parking Facilities*", Sustainable Energy, Grids and Networks, vol. 46, art. no. 102205, pp. 1-22, June 2026, DOI: 10.1016/j.segan.2026.102205, URL: http://doi.org/10.1016/j.segan.2026.102205. IMPACT FACTOR: 5.6.

[J5] V. Kratky, **G. Silano**, M. Vrba, C. Papaioannidis, I. Mademlis, R. Penicka, I. Pitas, and M. Saska. "*Gesture-Controlled Aerial Robot Formation for Human-Swarm Interaction in Safety Monitoring Applications*", IEEE Robotics and Automation Letters, vol. 10, no. 8, pp. 8244-8251, August 2025, DOI: 10.1109/LRA.2025.3583607, URL: http://doi.org/10.1109/LRA.2025.3583607. IMPACT FACTOR: 5.3.

[J6] **G. Silano**, A. Caballero, D. Liuzza, L. Iannelli, S. Bogdan, and M. Saska. "*A Signal Temporal Logic Approach for Task-Based Coordination of Multi-Aerial Systems: a Wind Turbine Inspection Case Study*", Robotics and Autonomous Systems, vol. 186, art. no. 104905, pp. 1-14, April 2025, DOI: 10.1016/j.robot.2024.104905, URL: http://doi.org/10.1016/j.robot.2024.104905. IMPACT FACTOR: 4.3.

[J7] D. Bonilla Licea, **G. Silano**, H. El Hammouti, M. Ghogho, and M. Saska. "*Reshaping UAV-Enabled Communications with Omnidirectional Multi-Rotor Aerial Vehicles*", IEEE Communications Magazine, vol. 63, no. 5, pp. 94-100, 2025, DOI: 10.1109/MCOM.001.2400421, URL: http://doi.org/10.1109/MCOM.001.2400421. IMPACT FACTOR: 8.3.

[J8] C. A. Dimmig, **G. Silano**, K. McGuire, C. Gabellieri, W. Hönig, J. Moore, and M. Kobilarov. "*Survey of Simulators for Aerial Robots: An Overview and In-Depth Systematic Comparisons*", IEEE Robotics and Automation Magazine, vol. 32, no. 2, pp. 153-166, June 2025, DOI: 10.1109/MRA.2024.3433171, URL: http://doi.org/10.1109/MRA.2024.3433171. IMPACT FACTOR: 7.1.

[J9] A. Caballero and **G. Silano**. "*A Signal Temporal Logic Motion Planner for Bird Diverter Installation Tasks with Multi-Robot Aerial Systems*", IEEE Access, vol. 11, pp. 81361-81377, 2023, DOI: 10.1109/ACCESS.2023.3300240, URL: http://doi.org/10.1109/ACCESS.2023.3300240. IMPACT FACTOR: 3.9.

[J10] D. Hert, T. Baca, P. Petracek, V. Kratky, R. Penicka, V. Spurny, M. Petrlik, M. Vrba, D. Zaitlik, P. Stoudek, V. Walter, P. Stepan, J. Horyna, V. Pritzl, M. Sramek, A. Ahmad, **G. Silano**, D. Bonilla Licea, P. Stibinger, T. Nascimento, and M. Saska. "*MRS Drone: A Modular Platform for Real-World Deployment of Aerial Multi-Robot Systems*", Journal of Intelligent & Robotic Systems, vol. 108, no. 64, pp. 1-34, 2023, DOI: 10.1007/s10846-023-01879-2, URL: https://doi.org/10.1007/s10846-023-01879-2. IMPACT FACTOR: 3.3.

[J11] A. Ahmad, D. Bonilla Licea, **G. Silano**, T. Baca, and M. Saska. "*PACNav: A Collective Navigation Approach for UAV Swarms Deprived of Communication and External Localization*", Bioinspiration & Biomimetics, vol. 17, no. 6, pp. 1-19, 2022, DOI: 10.1088/1748-3190/ac98e6, URL: https://doi.org/10.1088/1748-3190/ac98e6. IMPACT FACTOR: 2.985.

[J12] A. Basiri, V. Mariani, **G. Silano**, M. Aatif, L. Iannelli, and L. Glielmo. "*A survey on the application of path-planning algorithms for multi-rotor UAVs in precision agriculture*", The Journal of Navigation, vol. 75, no. 2, pp. 363-384, 2022, DOI: 10.1017/S0373463321000825, URL: https://doi.org/10.1017/S0373463321000825. IMPACT FACTOR: 1.995.

[J13] **G. Silano**, T. Baca, R. Penicka, D. Liuzza, and M. Saska. "*Power Line Inspection Tasks with Multi-Aerial Robot Systems via Signal Temporal Logic Specifications*", IEEE Robotics and Automation Letters, vol. 6, no. 2, pp. 4169-4176, 2021, DOI: 10.1109/LRA.2021.3068114, URL: http://dx.doi.org/10.1109/LRA.2021.3068114. IMPACT FACTOR: 3.608. ACCEPTED ALSO TO ICRA'21.

[J14] **G. Silano** and L. Iannelli. "*MAT-Fly: An Educational Platform for Simulating Unmanned Aerial Vehicles Aimed to Detect and Track Moving Objects*", IEEE Access, vol. 9, pp. 39333–39343, 2021, DOI: 10.1109/ACCESS.2021.3064758, URL: http://dx.doi.org/10.1109/ACCESS.2021.3064758. IMPACT FACTOR: 3.745.

Referred Book Chapters

[BC1] **G. Silano**, A. Kontou, Z. Feng, A. Acosta, O. Gehrke, T. A. Zerihun, S. Sanchez-Acevedo, G. Arnold, J. E. Rodriguez-Seco, R. Lazzari, C. Rodio, and L. Pellegrino. "*Experiences with Smart System Integration and Validation*". In European Guide to Smart Energy System Testing: The ERIGrid 2.0 Approach for Evaluating Complex Smart Energy System Configurations, Ed. by Thomas I. Strasser, Mihai Calin, Leonard Enrique Ramos Perez, SpringerBriefs in Energy, 2026, pp. 87-103, ISBN: 978-3-031-99451-7, DOI: 10.1007/978-3-031-99451-7_8, URL: https://doi.org/10.1007/978-3-031-99451-7_8.

[BC2] E. Widl, **G. Silano**, O. Gehrke, and T. A. Zerihun. "*Sector Coupling and Multi-Domain Systems Validation*". In European Guide to Smart Energy System Testing: The ERIGrid 2.0 Approach for Evaluating Complex Smart Energy System Configurations, Ed. by Thomas I. Strasser, Mihai Calin, Leonard Enrique Ramos Perez, SpringerBriefs in Energy, 2026, pp. 77-86, ISBN: 978-3-031-99451-7, DOI: 10.1007/978-3-031-99451-7_7, URL: https://doi.org/10.1007/978-3-031-99451-7_7.

[BC3] A. Acosta, **G. Silano**, G. Paludetto, O. Gehrke, M. C. Pham, Q. T. Tran, V. Rajkumar, S. Vogel, and A. Monti. "*Laboratory Infrastructure Integration and Automation*". In European Guide to Smart Energy System Testing: The ERIGrid 2.0 Approach for Evaluating Complex Smart Energy System Configurations, Ed. by Thomas I. Strasser, Mihai Calin, Leonard Enrique Ramos Perez, SpringerBriefs in Energy, 2026, pp. 65-76, ISBN: 978-3-031-99451-7, DOI: 10.1007/978-3-031-99451-7_6, URL: https://doi.org/10.1007/978-3-031-99451-7_6.

[BC4] **G. Silano** and L. Iannelli. "*CrazyS: a software-in-the-loop simulation platform for the Crazyflie 2.0 nano-quadcopter*". In Robot Operating System (ROS): The Complete Reference (Volume 4), Ed. by Koubaa, Anis, Springer International Publishing, 2020, pp. 81–115, ISBN: 978-3-030-20190-6, DOI: 10.1007/978-3-030-20190-6_4, URL: https://doi.org/10.1007/978-3-030-20190-6_4.

Peer-Reviewed Conference Papers

[C1] M. Bajja, A. K. A. H. Saliah, H. El Hammouti, D. Bonilla Licea, and **G. Silano**. "*3D Euler-Angle Orientation Control for Two-Ray Fading Mitigation in Maritime Air-to-Sea Communications*", 2026 IEEE Global Communications Conference (IEEE GLOBECOM). Macau S.A.R., China, December 2026, to appear.

[C2] **G. Silano**, Q. Sablé, M. Tognon, L. Iannelli, and A. Franchi. "*Sensitivity-Based Tube NMPC for Cooperative Aerial Structures Under Parametric Uncertainty*", 2026 International Conference on Unmanned Aircraft Systems (ICUAS). Corfu, Greece, June 2026, pp. 73-79, ISBN: 979-8-3315-9316-2, DOI: 10.1109/ICUAS69441.2026.11598699, URL: https://doi.org/10.1109/ICUAS69441.2026.11598699.

[C3] **G. Silano**, D. Bonilla Licea, H. El Hammouti, and M. Saska. "*Free-Space Optical Communication-Driven NMPC Framework for Multi-Rotor Aerial Vehicles in Structured Inspection Scenarios*", 2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC). Vienna, Austria, October 2025, pp. 6641-6646, ISBN: 979-8-3315-3358-8, DOI: 10.1109/SMC58881.2025.11343117, URL: https://doi.org/10.1109/SMC58881.2025.11343117.

[C4] D. Bonilla Licea, **G. Silano**, and M. Saska. "*Enhanced Security and Coordination Framework for UAV Swarms Using Heterogeneous Communication Networks*", In: Andreoni, M., Thakkar, S. (eds) Proceedings of the 1st GENZERO Workshop. GENZERO 2024, Hilton Yas, Abu Dhabi. Springer, Singapore, pp. 117–123, 2026, ISBN: 978-981-95-1050-4, DOI: 10.1007/978-981-95-1050-4_14, URL: https://doi.org/10.1007/978-981-95-1050-4_14.

[C5] D. Bonilla Licea, H. El Hammouti, **G. Silano**, and M. Saska. "*Harnessing the Potential of Omnidirectional Multi-Rotor Aerial Vehicles in Cooperative Jamming Against Eavesdropping*", 2024 IEEE Global Communications Conference (IEEE GLOBECOM). Cape Town, South Africa, December 2024, pp. 2052-2058, ISBN: 979-8-3503-5125-5, DOI: 10.1109/GLOBECOM52923.2024.10901802, URL: https://doi.org/10.1109/GLOBECOM52923.2024.10901802.

[C6] **G. Silano**, E. Rikos, V. Rajkumar, O. Gehrke, T. A. Zerihun, C. Rodio, and R. Lazzari. "*Integrating Power-to-Heat Services in Geographically Distributed Multi-Energy Systems: A Case Study from the ERIGrid 2.0 Project*", 2024 Open Source Modelling and Simulation of Energy Systems (OSMSES 2024). Vienna, Austria, September 2024, pp. 1-5, ISBN: 979-8-3503-8468-0, DOI: 10.1109/OSMSES62085.2024.10668976, URL: https://doi.org/10.1109/OSMSES62085.2024.10668976.

[C7] V. Rajkumar, **G. Silano**, O. Gehrke, S. Vogel, E. Widl, G. Paludetto, E. Rikos, T. A. Zerihun, A. Stefanov, P. Palensky, and T. I. Strasser. "*Laboratory Middleware for the Cyber-Physical Integration of Energy Research Infrastructures*", 2024 12th Workshop on Modelling and Simulation of Cyber-Physical Energy Systems (MSCPES). Hong Kong, China, May 2024, pp. 1-5, ISBN: 979-8-3503-6284-8, DOI: 10.1109/MSCPES62135.2024.10542755, URL: https://doi.org/10.1109/MSCPES62135.2024.10542755.

[C8] D. Bonilla Licea, **G. Silano**, M. Ghogho, and M. Saska. "*Omnidirectional Multi-Rotor Aerial Vehicle Pose Optimization: A Novel Approach to Physical Layer Security*", 2024 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP). Seoul, Korea, April 2024, pp. 9021-9025, ISBN: 979-8-3503-4485-1, DOI: 10.1109/ICASSP48485.2024.10447876, URL: https://doi.org/10.1109/ICASSP48485.2024.10447876.

[C9] **G. Silano**, A. Afifi, M. Saska, and A. Franchi. "*A Signal Temporal Logic Planner for Ergonomic Human–Robot Collaboration*", 2023 International Conference on Unmanned Aircraft Systems (ICUAS). Lazarski University, Warsaw, Poland, June 2023, pp. 328-335, ISBN: 979-8-3503-1037-5, DOI: 10.1109/ICUAS57906.2023.10156559, URL: https://doi.org/10.1109/ICUAS57906.2023.10156559.

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[W7] **G. Silano**, V. Kratky, M. Vrba, C. Papaioannidis, I. Mademlis, R. Penicka, I. Pitas, and M. Saska. "*Human-Swarm Interaction with a Gesture-Controlled Aerial Robot Formation for Safety Monitoring Applications*". IROS 2023 Human Multi-Robot Interaction Workshop, Detroit, USA, October 2023, pp. 1-5, URL: https://mrs.felk.cvut.cz/data/papers/iros2023_hsi_ws.pdf.

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[W10] **G. Silano**, D. Liuzza, L. Iannelli, and M. Saska. "*A framework for power line inspection tasks with multi-robot systems from signal temporal logic specifications*". SIDRA[6] National Meeting, Cagliari, Italy, September 2020, pp. 1-2, DOI: 10.48550/arXiv.2103.02999, URL: https://arxiv.org/abs/2103.02999.

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Posters

[PO1] **G. Silano**, P. Oppido, and L. Iannelli. "*Software-in-the-loop simulation for improving flight control system design: a quadrotor case study*". In SIDRA[6] National Meeting, Ancona, Italy, September 2019, DOI: 10.13140/RG.2.2.31583.61603, URL: http://dx.doi.org/10.13140/RG.2.2.31583.61603.

[PO2] **G. Silano** and L. Iannelli. "*An educational simulation platform for Unmanned Aerial Vehicles aimed to detect and track moving objects*". In SIDRA[6] National Meeting, Milan, Italy, September 2017, DOI: 10.13140/RG.2.2.14878.43849, URL: http://dx.doi.org/10.13140/RG.2.2.14878.43849.

Miscellaneous

[M1] T. Prihodova, **G. Silano**, A. Ahmad, V. Kratky, T. Baca, P. Petracek, V. Saskova, J. Bednar, and M. Saska. "*2022 IEEE Robotics and Automation Society Summer School on Multi-Robot Systems in Prague [Education]*", IEEE Robotics and Automation Magazine, vol. 30, no. 1, pp. 104–106, 2023, DOI: 10.1109/MRA.2023.3238213, URL: https://doi.org/10.1109/MRA.2023.3238213. IMPACT FACTOR: 5.229.

Theses

Dec 2016 – Jul 2020 **Ph.D. Thesis**, *Software-in-the-loop methodologies for the analysis and control design of small UAV systems*, University of Sannio, Benevento, Italy

- Showed the role and the effectiveness of robotic simulators and Software-in-the-loop methodologies in the flight control system design of multi-rotor vehicles.
- Advisor: Prof. Dr. Luigi Iannelli.

Sep 2015 – Mar 2016 **Master Thesis**, *Development of a simulator aimed to detect and track moving objects for UAVs*, University of Sannio, Benevento, Italy

- Design and development of a 3D simulator aimed to detect and track arbitrary moving objects for UAVs.
- Advisor: Prof. Dr. Luigi Iannelli.

Feb 2012 – Jun 2012 **Bachelor Thesis**, *An Object Oriented approach aimed to the creation of control software for industrial processes*, Mosaico Monitoraggio Integrato S.r.l., Benevento, Italy

- Design of a software production methodology using an Object Oriented (OO) approach for Programmable Logic Controllers (PLCs).
- Advisors: Prof. Dr. Luigi Iannelli and Eng. Paolo Rubino.